THE DIMENSIONS OF THE FIBRES OF A PSEUDO-COHERENT COMPLEX ARE UPPER SEMI-CONTINUOUS WITH CONSTRUCTIBLE LEVEL SETS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The derived pullback of the residue field at a point of the source agrees with the pullback of the residue field at the image, which gives the base change compatibility of the dimension functions. used at proof lines 7924-7938. Proves the first assertion of the statement.
- A pseudo-coherent complex can locally be represented by a bounded above complex of finite free modules, and cutting such a complex at a degree gives a perfect complex with the same top cohomology. invoked at proof lines 7940-7948. Reduces the remaining assertions to the case of a perfect complex.
- A perfect complex is locally represented by a bounded complex of finite free modules with the ranks prescribed by the dimensions of the fibres. invoked at proof lines 7949-7958. Turns the problem into a statement about the ranks of the modules in a bounded complex, i.e. into algebra over a ring.
- For a bounded complex of finite free modules over a ring the dimension functions are semicontinuous and their level sets are constructible. used at proof lines 7959-7992. Supplies the algebra input that finishes the semicontinuity and the constructibility.

Source reference. Original lemma 08EE: sources/stacks-perfect/pinned-original-source.tex, source lines 3321–3321. Invoked at proof lines 7940-7942 to represent the complex on an affine neighbourhood; the truncation reducing the problem to the perfect case is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 7956-7958 in support of turning the problem into algebra; the reduction steps that precede it are inside the counted span.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Cited with the previous label at proof lines

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7956-7958 for the same reduction; the algebra conclusion is inside the counted span.

Source reference. Original lemma 0BCD: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 7949-7955 to represent the perfect complex by a bounded complex of finite free modules with the ranks given by the dimensions at the point; the conclusion drawn is inside the counted span.

Source reference. Original lemma 0BCJ: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer in the reduction to algebra; the reduction and the conclusion are inside the counted span.

Lemma 0.2. *Let X be a scheme. Let $E \in D(\mathcal{O}_X)$ be pseudo-coherent (for example perfect). For any $i \in \mathbf{Z}$ consider the function*

$$\beta_i : X \longrightarrow \{0, 1, 2, \ldots\}, \quad x \longmapsto \dim_{\kappa(x)} H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \kappa(x))$$

Then we have

1. *formation of β_i commutes with arbitrary base change,*
2. *the functions β_i are upper semi-continuous, and*
3. *the level sets of β_i are locally constructible in X.*

Proof. Consider a morphism of schemes $f : Y \to X$ and a point $y \in Y$. Let x be the image of y and consider the commutative diagram

$$\begin{array}{ccc} y & \xrightarrow[j]{} & Y \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ x & \xrightarrow{i} & X \end{array}$$

Then we see that $Lg^* \circ Li^* = Lj^* \circ Lf^*$. This implies that the function β'_i associated to the pseudo-coherent complex Lf^*E is the pullback of the function β_i, in a formula: $\beta'_i = \beta_i \circ f$. This is the meaning of (1).

Fix i and let $x \in X$. It is enough to prove (2) and (3) holds in an open neighbourhood of x, hence we may assume X affine. Then we can represent E by a bounded above complex $\mathcal{F}^\bullet$ of finite free modules (Lemma 08EE). Then $P = \sigma_{\geq i-1}\mathcal{F}^\bullet$ is a perfect object and $P \to E$ induces an isomorphism

$$H^i(P \otimes^{\mathbf{L}}_{\mathcal{O}_X} \kappa(x')) \to H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \kappa(x'))$$

for all $x' \in X$. Thus we may assume E is perfect. In this case by More on Algebra, Lemma 0BCD there exists an affine open neighbourhood U of x and $a \leq b$ such that $E|_U$ is represented by a complex

$$\ldots \to 0 \to \mathcal{O}_U^{\oplus \beta_a(x)} \to \mathcal{O}_U^{\oplus \beta_{a+1}(x)} \to \ldots \to \mathcal{O}_U^{\oplus \beta_{b-1}(x)} \to \mathcal{O}_U^{\oplus \beta_b(x)} \to 0 \to \ldots$$

(This also uses earlier results to turn the problem into algebra, for example Lemmas 06Z0 and 08EB.) It follows immediately that $\beta_i(x') \leq \beta_i(x)$ for all $x' \in U$. This proves that β_i is upper semi-continuous.

To prove (3) we may assume that X is affine and E is given by a complex of finite free $\mathcal{O}_X$-modules (for example by arguing as in the previous paragraph, or by using Cohomology, Lemma 0BCJ). Thus we have to show that given a complex

$$\mathcal{O}_X^{\oplus a} \to \mathcal{O}_X^{\oplus b} \to \mathcal{O}_X^{\oplus c}$$

the function associated to a point $x \in X$ the dimension of the cohomology of $\kappa_x^{\oplus a} \to \kappa_x^{\oplus b} \to \kappa_x^{\oplus c}$ in the middle has constructible level sets. Let $A \in \mathrm{Mat}(a \times b, \Gamma(X, \mathcal{O}_X))$ be the matrix of the first arrow. The rank of the image of A in $\mathrm{Mat}(a \times b, \kappa(x))$ is equal to r if all $(r+1) \times (r+1)$-minors of A vanish at x and there is some $r \times r$-minor of A which does not vanish at x. Thus the set of points where the rank is r is a constructible locally closed set. Arguing similarly for the second arrow and putting everything together we obtain the desired result. $\square$